روافد
RAWAAF

Software Requirements Specification (SRS)

Smart short-stay booking & operations platform — Guest website/app, Owner & Operator dashboard, Field apps, Platform Admin

Item	Value
Document	Software Requirements Specification
Product	Rawaaf (روافد) — Smart Stay Platform
Version	1.0 — Draft for client review
Date	29 September 2026
Status	For review & sign-off
Companion deliverables	Clickable UI/UX prototype (prototype/index.html); Product vision presentation (docs/business-experience)

Document Control

Revision history

Version	Date	Author	Changes
0.1	22 Sep 2026	Business Analysis	Product vision & business-analysis presentation (Arabic).
1.0	29 September 2026	Business Analysis	First full SRS. Consolidates client vision + client Rawaaf prototype; adds NFRs, data model, interfaces, open questions.

How to read requirement tags

Every requirement carries a Priority, a Status and a Source so that assumptions are never read as decisions.

Tag	Values	Meaning
Priority (MoSCoW)	Must / Should / Could / Won't	Must = required for launch (MVP). Should = important, can follow shortly. Could = nice to have. Won't = out of current scope.
Status	Confirmed	Stated explicitly by the client (vision or prototype).
	Proposed	Capability proposed by the analyst team to make the product complete.
	Recommended	Enhancement that raises quality or reduces risk (security, compliance, operations).
	TBC	Needs a client decision or information before design — see Section 10.
	Future	Outside initial scope.
Source	Client vision / Client prototype / Analyst	Where the requirement came from, for traceability.

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of Rawaaf (رواف), a smart short-stay platform in the Kingdom of Saudi Arabia that combines property booking, smart (keyless) access, property operations and supervision, AI-assisted pricing, a personal guest assistant, loyalty & wallet, KPI dashboards, advertising campaigns and government compliance integrations (SHOMOOS, NAFATH, EWAA).

It is the agreed reference for the client, designers, developers and testers. It is written to be approved by the client, and every open decision is listed explicitly in Section 10.

1.2 Scope

In scope:

- Guest web experience (responsive), and the same features for a mobile app if confirmed (OQ-BUS-01).
- Owner & Operator dashboard; Supervisor and Photographer field apps (mobile-first web).
- Platform Admin console.
- Integrations: payment gateway, smart-lock provider, SHOMOOS, NAFATH, EWAA, SMS/WhatsApp/e-mail, maps, AI provider, ZATCA e-invoicing.

Out of scope for this version: demand forecasting, personalised recommendations, proactive operational insights (Future), channel-manager sync with OTAs (Airbnb/Booking.com) unless requested, and native hardware installation of locks.

1.3 Product vision

“The smarter way to operate and experience short-term stays.” Eight systems, one experience:

- Property booking
- Smart access
- Property operations
- Supervision
- Pricing intelligence
- Compliance integrations
- Personal assistance
- Business analytics

Together they serve three audiences: guests, property owners/operators, and platform management.

1.4 Definitions & acronyms

Term	Definition
ADR	Average Daily Rate.
RevPAR	Revenue per Available Room/unit-night.
ROAS	Return on Advertising Spend.
OTP	One-Time Password.
RBAC	Role-Based Access Control.
PDPL	Saudi Personal Data Protection Law.

Term	Definition
ZATCA	Zakat, Tax and Customs Authority (e-invoicing “Fatoora”).
SHOMOOS (شموس)	Government system for registering lodger data in accommodation facilities (expected purpose).
NAFATH (نفاذ)	National Single Sign-On / digital identity verification.
EWAA (إيواء)	Accommodation-sector integration named by the client; entity and scope TBC.
Supervised property	Property served by Rawaaf field team (supervisor, photographer, field support).
Self-service property	Property operated fully digitally without field presence.
Digital key / Access credential	PIN or e-key bound to a booking that opens the smart lock.
Wallet	Rawaaf stored-value balance in SAR used for payments, cashback and rewards.
Cashback	Percentage of a paid booking credited back to the guest wallet.
Superhost (مضيف مميز)	Badge for high-performing hosts.
MVP	Minimum Viable Product.
TBC	To Be Confirmed by the client.

1.5 References

- Rawaaf product vision & business-analysis presentation — docs/business-experience/index.html (Sep 2026).
- Client prototype “Rawaaf client account” web app — docs/client-reference/ (shared by client via Meta AI).
- Rawaaf clickable UI/UX prototype — prototype/index.html (companion to this SRS).
- ISO/IEC/IEEE 29148:2018 — Requirements engineering (structure reference).
- Saudi Personal Data Protection Law (PDPL) and implementing regulations; ZATCA e-invoicing regulations.

2. Overall Description

2.1 Product perspective

Rawaaf is a new, self-contained multi-sided platform. It sits between guests, property owners/operators, a field team, smart locks installed in units, and external government and commercial services. The core data flow is:

Guest → Booking → Reservation data → Access credential → Smart lock → Property → Operational data → Analytics → KPI dashboard & Pricing assistant

From reservation data, compliance data flows to SHOMOOS / NAFATH / EWAA (details TBC).

2.2 Product functions (modules)

§	Code	Module	Summary
3.1	USR	Identity, Accounts & Access Control	Registration, authentication, identity verification and role-based access for all human users of the platform.
3.2	DSC	Discovery & Search	How guests find units: browsing, searching, filtering, favourites and comparison.
3.3	PDP	Property Detail & Reviews	The unit detail page and the verified review system.
3.4	BKG	Booking & Availability	The booking wizard, availability control, booking lifecycle, modification and cancellation.
3.5	PAY	Payments, Wallet & Invoicing	Payment methods, the Rawaaf digital wallet, cashback, refunds and tax invoices.
3.6	LOY	Loyalty, Points & Referrals	Points, tiers, rewards and referral programme shown in the client prototype.
3.7	ACC	Smart Access	Digital key issuance, smart-lock integration, access logging and fallback.
3.8	STY	Guest Stay Experience	Everything the guest needs from arrival to departure.
3.9	AST	Personal AI Assistant	A conversational assistant that accompanies the guest across the journey.
3.10	PRP	Property & Unit Management	Onboarding and managing properties and units by Owners and Operators.
3.11	OPS	Operations & Supervision	Field operations for supervised properties: readiness, inspections, tasks and incidents.
3.12	MED	Content & Photography	Professional photography workflow and content quality review.
3.13	PRC	Smart Pricing	AI pricing assistant that recommends nightly prices based on demand signals.
3.14	KPI	KPI Dashboards & Reporting	Performance dashboards for Admin, Owners and Operators.
3.15	CMP	Campaigns & Promotions	Advertising campaigns and promotions with conversion tracking.
3.16	INT	Compliance & Government Integrations	Integrations with SHOMOOS, NAFATH and EWAA. Exact data contracts are TBC.
3.17	POL	Policies, Damage & Disputes	Damage & liability policy enforcement, incident review and fines.
3.18	NTF	Notifications	Transactional notifications across channels.
3.19	FIN	Financials & Payouts	Revenue split between platform and owners, statements and payouts.
3.20	ADM	Platform Administration	Back-office configuration and oversight.

2.3 User classes & characteristics

Role	Description	Data scope	Primary channel
Guest	Discovers, books and stays in units; uses wallet, points, digital key and assistant.	Own account, bookings, wallet, reviews.	Web (responsive), mobile app (TBC).
Property Owner	Owns properties; lists them, approves policies, follows performance and payouts.	Own properties only.	Owner dashboard (web).
Property Operator	Runs day-to-day operations: calendar, pricing, bookings, readiness, access settings.	Assigned properties only.	Operator dashboard (web).
Supervisor	Field staff for supervised properties: readiness checks, post-stay inspection, incident reporting.	Tasks of assigned properties.	Mobile-first field app.
Photographer	Takes professional photos for supervised properties and submits for QA.	Assigned shoots only.	Mobile-first field app.
Platform Admin	Manages the whole platform: users, verification, policies, campaigns, integrations, finance.	Full platform scope (sub-roles recommended: Super Admin, Operations, Finance, Support, Content).	Admin console (web).
Integration (system)	Non-human actor exchanging data with SHOMOOS, NAFATH, EWAA, payment, lock and messaging providers.	Specific API scopes only.	Server-to-server APIs.

Not every role necessarily needs a separate login (e.g. Photographer) — to be confirmed.

2.4 Operating models

Aspect	Supervised	Self-service
Field presence	Rawaaf supervisor, photographer, field support	None — fully digital
Content	Professional photography + QA	Owner uploads photos; QA by admin
Unit readiness	Supervisor readiness inspection before each arrival	Owner/service provider responsibility (TBC)
Arrival	Smart access, optional field support	Smart access only
During stay	Field support + personal assistant	Personal assistant + remote support
After departure	Supervisor post-stay inspection with photos	Owner/provider inspection (TBC)

2.5 Operating environment

- Cloud-hosted, with data residency in Saudi Arabia (recommended: a KSA cloud region).
- Guest: modern mobile & desktop browsers; optional iOS/Android app.
- Field apps: mobile browsers (installable PWA), camera access for photos.
- Smart locks: provider cloud API (Wi-Fi/gateway) and/or Bluetooth with offline codes.

2.6 Design & implementation constraints

- Arabic-first RTL UI following the Rawaaf visual identity supplied by the client (Section 6.1).
- Regulatory: PDPL, ZATCA e-invoicing, CST SMS rules, Ministry of Tourism licensing for short-term rentals, SHOMOOS obligations.
- Payment through a SAMA-licensed gateway; no card data stored by the platform.
- Smart-lock capabilities limited by the chosen provider’s API.

2.7 Assumptions & dependencies

- The client will provide access to government integration channels (SHOMOOS, NAFATH, EWAA) and the related agreements.
- The client will contract a payment gateway, SMS/WhatsApp provider and smart-lock provider.
- Units in the Smart-access programme will have compatible locks installed before go-live.
- Prices, fees, points and cashback figures in the client prototype are treated as confirmed business rules unless changed in Section 10.

3. Functional Requirements

Requirements are grouped by module. IDs follow FR-<MODULE>-<NN> and are stable for traceability to design, prototype screens and test cases.

3.1 Identity, Accounts & Access Control (USR)

Registration, authentication, identity verification and role-based access for all human users of the platform.

ID	Requirement	Priority	Status	Source
FR-USR-01	The system shall allow a guest to register using a Saudi mobile number (format 05XXXXXXXX, 10 digits), full name, optional e-mail and password.	Must	Confirmed	Client prototype
FR-USR-02	The system shall verify mobile-number ownership with a one-time password (OTP) delivered by SMS before activating an account.	Must	Recommended	Analyst
FR-USR-03	The system shall allow users to sign in with mobile number and password, or mobile number and OTP.	Must	Proposed	Client prototype
FR-USR-04	The system shall award 100 welcome loyalty points on successful registration.	Should	Confirmed	Client prototype
FR-USR-05	The system shall support identity verification through Nafath (National Single Sign-On) at the point defined in OQ-INT-02.	Must	TBC	Client vision
FR-USR-06	The system shall support the following roles: Guest, Property Owner, Property Operator, Supervisor, Photographer, Platform Admin, and a non-human Integration account.	Must	Confirmed	Client vision
FR-USR-07	The system shall enforce role-based access control (RBAC) so that each role only sees the data scoped to it (e.g. an Owner sees only their properties; an Operator only assigned properties).	Must	Proposed	Client vision
FR-USR-08	The Platform Admin shall be able to invite, suspend, reactivate and delete users and change their roles.	Must	Proposed	Client vision
FR-USR-09	The system shall allow a user to edit profile data (name, mobile, e-mail) and notification preferences.	Must	Confirmed	Client prototype
FR-USR-10	The system shall allow a guest to permanently delete their account after explicit confirmation, subject to legal retention rules for bookings, invoices and compliance records.	Must	Confirmed	Client prototype
FR-USR-11	The system shall lock an account for 15 minutes after 5 consecutive failed sign-in attempts.	Should	Recommended	Analyst
FR-USR-12	The system shall support two-factor authentication for Owner, Operator and Admin roles.	Should	Recommended	Analyst

3.2 Discovery & Search (DSC)

How guests find units: browsing, searching, filtering, favourites and comparison.

ID	Requirement	Priority	Status	Source
FR-DSC-01	The system shall allow guests to search available units by free text (city, district, unit name).	Must	Confirmed	Client vision + Client prototype
FR-DSC-02	The system shall support property types: apartment, hotel (room/suite), residential unit, villa, chalet, rest house (istiraha) and mountain cabin.	Must	Confirmed	Client vision + Client prototype

ID	Requirement	Priority	Status	Source
FR-DSC-03	The system shall allow filtering by city, property type, check-in/check-out dates, number of guests, price range, amenities, operating model (supervised / self-service) and smart-access availability.	Must	Proposed	Client vision + Client prototype
FR-DSC-04	The system shall exclude units that are unavailable for the selected dates from date-based search results.	Must	Confirmed	Client vision
FR-DSC-05	The system shall display each unit card with main photo, title, city and district, rating and review count, price per night, original price and discount % (if any), “Superhost” badge and “Smart access” badge.	Must	Confirmed	Client prototype
FR-DSC-06	The system shall allow a signed-in guest to add/remove units to Favourites; unauthenticated users are prompted to sign in.	Must	Confirmed	Client prototype
FR-DSC-07	The system shall allow a guest to select up to 3 units and view a side-by-side comparison (price, capacity, bedrooms, rating, amenities, policies).	Should	Confirmed	Client prototype
FR-DSC-08	The system shall allow sorting results by recommended, price (low/high), rating and newest.	Should	Proposed	Analyst
FR-DSC-09	The system shall show results on a map view.	Could	Proposed	Analyst

3.3 Property Detail & Reviews (PDP)

The unit detail page and the verified review system.

ID	Requirement	Priority	Status	Source
FR-PDP-01	The system shall display a unit detail page with photo gallery, description, capacity (guests, bedrooms), amenities, location, host information, house rules, cancellation policy and damage policy.	Must	Confirmed	Client vision + Client prototype
FR-PDP-02	The system shall display rating summary: average rating, total reviews and distribution by stars (5 to 1).	Must	Confirmed	Client prototype
FR-PDP-03	The system shall allow reviews to be sorted by newest, highest rating and “with photos”.	Should	Confirmed	Client prototype
FR-PDP-04	The system shall only allow a guest to review a unit after a booking for that unit reaches status Completed (verified reviews).	Must	Confirmed	Client prototype
FR-PDP-05	The system shall allow a review to include a 1–5 star rating, text comment and optional photos.	Must	Confirmed	Client prototype
FR-PDP-06	The system shall award 50 loyalty points for each published review.	Should	Confirmed	Client prototype
FR-PDP-07	The system shall allow the guest to edit or delete their own review.	Should	Confirmed	Client prototype
FR-PDP-08	The system shall allow the Owner/Operator to post one public reply per review.	Should	Confirmed	Client prototype
FR-PDP-09	The system shall allow a guest to subscribe to a price-drop alert for a unit and notify them when the nightly price decreases.	Could	Confirmed	Client prototype
FR-PDP-10	The system shall allow sharing a unit link and contacting the host/support via WhatsApp.	Should	Confirmed	Client prototype
FR-PDP-11	The Admin shall be able to hide reviews that violate content policy, with a logged reason.	Should	Recommended	Analyst

3.4 Booking & Availability (BKG)

The booking wizard, availability control, booking lifecycle, modification and cancellation.

ID	Requirement	Priority	Status	Source
FR-BKG-01	The system shall check real-time availability of a unit for the selected dates before allowing a booking.	Must	Confirmed	Client vision
FR-BKG-02	The system shall prevent double booking of the same unit for overlapping nights, including under concurrent requests.	Must	Proposed	Client vision
FR-BKG-03	The system shall provide a booking wizard: (1) dates & guests, (2) guest details & identity verification, (3) policy acknowledgement, (4) payment, (5) confirmation.	Must	Proposed	Client prototype
FR-BKG-04	The system shall show a full price breakdown before confirmation: nightly rate × nights, cleaning fee, discounts, VAT (15%), total, and expected cashback and points.	Must	Proposed	Client vision + Client prototype
FR-BKG-05	The system shall limit the number of guests to the unit's maximum capacity.	Must	Confirmed	Client prototype
FR-BKG-06	The system shall require the guest to acknowledge the property policy (house rules, damage and liability) before confirmation, and store the acknowledgement with date, time and policy version.	Must	Proposed	Client vision
FR-BKG-07	The system shall issue a booking confirmation with a unique booking number after successful payment.	Must	Confirmed	Client vision + Client prototype
FR-BKG-08	The system shall manage booking statuses: Pending Payment, Confirmed, Checked-in, Completed, Cancelled, No-show.	Must	Proposed	Analyst
FR-BKG-09	The system shall allow the guest to view all bookings with status, dates, total and payment method, and to open booking details.	Must	Confirmed	Client prototype
FR-BKG-10	The system shall allow the guest to cancel a confirmed booking and apply the unit's cancellation policy to compute the refund.	Must	TBC	Client vision + Client prototype
FR-BKG-11	The system shall allow the guest to request a date modification, subject to availability and the modification policy.	Should	TBC	Client vision
FR-BKG-12	The system shall allow the guest to "Book again" from a past booking.	Could	Confirmed	Client prototype
FR-BKG-13	The system shall hold selected dates for 10 minutes while payment is in progress and release them if payment is not completed.	Must	Recommended	Analyst
FR-BKG-14	The system shall allow Operators to block dates on a unit calendar (maintenance, owner use).	Must	Proposed	Client vision

3.5 Payments, Wallet & Invoicing (PAY)

Payment methods, the Rawaaf digital wallet, cashback, refunds and tax invoices.

ID	Requirement	Priority	Status	Source
FR-PAY-01	The system shall integrate with a licensed Saudi payment gateway supporting mada, Visa/Mastercard and Apple Pay (provider TBC).	Must	TBC	Client vision + Client prototype
FR-PAY-02	The system shall provide each guest with a digital wallet in SAR showing current balance.	Must	Confirmed	Client prototype

ID	Requirement	Priority	Status	Source
FR-PAY-03	The system shall allow a guest to top up the wallet with preset amounts (100, 300, 500 SAR) or a custom amount through the payment gateway.	Must	Confirmed	Client prototype
FR-PAY-04	The system shall allow paying a booking fully from the wallet; if the balance is insufficient it shall block the payment and offer top-up or another method.	Must	Confirmed	Client prototype
FR-PAY-05	The system shall credit cashback to the wallet after a paid booking at the rate of the guest's loyalty tier (default 5%).	Must	Confirmed	Client prototype
FR-PAY-06	The system shall record every wallet movement (top-up, booking payment, cashback, reward redemption, refund, referral bonus) in a transaction history with date, type, amount and description.	Must	Confirmed	Client prototype
FR-PAY-07	The system shall allow saving a tokenised card (never storing raw card data) and display only the last 4 digits.	Should	Confirmed	Client prototype
FR-PAY-08	The system shall issue a VAT-compliant electronic invoice for each paid booking, compliant with ZATCA e-invoicing (Fatoora) requirements, downloadable as PDF.	Must	Recommended	Client prototype + Analyst
FR-PAY-09	The system shall process refunds to the original payment method or to the wallet, according to the cancellation policy and admin decision.	Must	TBC	Analyst
FR-PAY-10	The system shall reverse cashback and points that were earned on a booking that is later cancelled.	Must	Recommended	Analyst
FR-PAY-11	The system shall treat wallet balance as non-expiring unless a different rule is approved.	Should	TBC	Client prototype

3.6 Loyalty, Points & Referrals (LOY)

Points, tiers, rewards and referral programme shown in the client prototype.

ID	Requirement	Priority	Status	Source
FR-LOY-01	The system shall award 10 points for every 100 SAR paid on a booking.	Must	Confirmed	Client prototype
FR-LOY-02	The system shall allow redeeming points in blocks of 100 points = 10 SAR credited instantly to the wallet.	Must	Confirmed	Client prototype
FR-LOY-03	The system shall assign loyalty tiers by points: Bronze (0–799), Silver (800–1,999), Gold (2,000+), and show progress to the next tier.	Must	Confirmed	Client prototype
FR-LOY-04	The system shall apply tier benefits: Bronze — 3% cashback, standard support; Silver — 5% cashback, priority support, cleaning discount; Gold — 7% cashback, account manager, free upgrade (subject to OQ-LOY-01).	Should	TBC	Client prototype
FR-LOY-05	The system shall generate a unique referral code per guest; when a referred user completes their first booking both parties receive 50 SAR in their wallets.	Should	Confirmed	Client prototype
FR-LOY-06	The system shall grant a free cleaning after every 5 non-cancelled bookings and show progress in the account.	Could	Confirmed	Client prototype
FR-LOY-07	The Admin shall be able to configure cashback %, points earn/burn rates, tier thresholds and benefits without code changes.	Must	Recommended	Analyst

3.7 Smart Access (ACC)

Digital key issuance, smart-lock integration, access logging and fallback.

ID	Requirement	Priority	Status	Source
FR-ACC-01	The system shall generate an access credential (digital key / PIN) for every confirmed booking.	Must	Confirmed	Client vision + Client prototype
FR-ACC-02	The system shall bind each credential to exactly one booking and one unit.	Must	Confirmed	Client vision
FR-ACC-03	The system shall deliver the credential to the guest in the app/website, and by SMS/WhatsApp, after confirmation.	Must	Confirmed	Client vision
FR-ACC-04	The system shall integrate with the selected smart-lock provider API to provision, update and revoke credentials on the lock (provider TBC).	Must	Confirmed	Client vision
FR-ACC-05	The credential shall be valid only from check-in time (default 15:00) on the arrival date until check-out time (default 12:00) on the departure date.	Must	TBC	Client vision + Client prototype
FR-ACC-06	The system shall mask the credential by default and let the guest reveal it; for Bluetooth/online locks it shall offer an “Open door” action.	Should	Proposed	Analyst
FR-ACC-07	The system shall log every access attempt (success and failure) with timestamp, unit, booking and method.	Must	Confirmed	Client vision
FR-ACC-08	The system shall automatically revoke the credential when a booking is cancelled, and update it when dates are modified.	Must	Recommended	Client vision
FR-ACC-09	The system shall shorten or extend validity when an early check-in / late check-out is approved.	Should	Proposed	Analyst
FR-ACC-10	The system shall provide a fallback procedure when the lock is offline or fails (e.g. offline codes generated by the lock algorithm, emergency code issued by Operator, or field support).	Must	Recommended	Client vision
FR-ACC-11	The system shall monitor lock health (online status, battery level) and alert Operators when battery < 20% or the lock is offline.	Should	Recommended	Analyst

3.8 Guest Stay Experience (STY)

Everything the guest needs from arrival to departure.

ID	Requirement	Priority	Status	Source
FR-STY-01	The system shall show an “Active stay” screen during the stay with digital key, Wi-Fi details, property guide, check-out time and support.	Must	Proposed	Client vision
FR-STY-02	The system shall send pre-arrival instructions (address, map link, access steps) 24 hours before check-in.	Must	Proposed	Client vision
FR-STY-03	The system shall allow the guest to request early check-in or late check-out; the Operator approves or rejects and any fee is charged.	Should	Proposed	Analyst
FR-STY-04	The system shall allow the guest to report an issue (category, description, photo) that creates a ticket for Operations.	Must	Proposed	Client vision
FR-STY-05	The system shall allow a self check-out action which closes the stay, expires the key and triggers post-stay inspection.	Should	Proposed	Client vision
FR-STY-06	The system shall send a review request after check-out.	Should	Confirmed	Client prototype

3.9 Personal AI Assistant (AST)

A conversational assistant that accompanies the guest across the journey.

ID	Requirement	Priority	Status	Source
FR-AST-01	The system shall provide a personal assistant chat to guests in Arabic and English.	Must	Confirmed	Client vision
FR-AST-02	The assistant shall answer questions about bookings, the property (from the property profile) and stay information.	Must	Recommended	Client vision
FR-AST-03	The assistant shall provide arrival guidance and smart-lock usage instructions, including troubleshooting steps.	Must	Recommended	Client vision
FR-AST-04	The assistant shall suggest suitable units for given dates and needs during discovery.	Should	Recommended	Client vision
FR-AST-05	The assistant shall provide local information (nearby pharmacies, restaurants, services) — data source TBC.	Could	TBC	Client vision
FR-AST-06	The assistant shall create support tickets and hand over to a human agent when it cannot resolve a request, or when the guest asks.	Must	Recommended	Client vision
FR-AST-07	The assistant shall only access data belonging to the signed-in guest's own bookings.	Must	Recommended	Analyst
FR-AST-08	The assistant shall not perform financial actions or modify/cancel bookings without explicit guest confirmation and an approved permission (OQ-AST-02).	Must	TBC	Client vision
FR-AST-09	The system shall log assistant conversations for quality review, subject to the privacy policy.	Should	Recommended	Analyst

3.10 Property & Unit Management (PRP)

Onboarding and managing properties and units by Owners and Operators.

ID	Requirement	Priority	Status	Source
FR-PRP-01	The system shall allow an Owner/Operator to add a property through an onboarding wizard: type, location (map pin), units, capacity, amenities, photos, policies, operating model and smart-lock setup.	Must	Confirmed	Client vision
FR-PRP-02	The system shall organise each property profile in five data layers: Identity, Content, Commercial, Operations and Governance.	Must	Proposed	Client vision
FR-PRP-03	The system shall allow managing photos (upload, reorder, set cover) and amenities.	Must	Confirmed	Client vision
FR-PRP-04	The system shall allow managing an availability calendar per unit.	Must	Confirmed	Client vision
FR-PRP-05	The system shall allow setting the operating model per property: Supervised or Self-service.	Must	Confirmed	Client vision
FR-PRP-06	The system shall allow configuring access settings per unit (lock provider, lock ID, test code).	Must	Confirmed	Client vision
FR-PRP-07	The system shall display the booking status and operational status of each unit (Available, Booked, Occupied, Preparing, Post-checkout inspection, Draft, Suspended).	Must	Confirmed	Client vision
FR-PRP-08	The system shall allow defining house rules, cancellation policy (from admin templates) and damage policy per property, and display them to guests.	Must	Confirmed	Client vision
FR-PRP-09	The system shall require verification of ownership or operator authorisation (documents such as title deed / lease and tourism licence) before a property can be published.	Must	TBC	Client vision

ID	Requirement	Priority	Status	Source
FR-PRP-10	The system shall not publish a property until mandatory data and minimum photos (e.g. 8) are complete and verified.	Must	Proposed	Client vision
FR-PRP-11	The system shall allow an Owner to assign one or more Operators to their properties.	Should	Proposed	Analyst

3.11 Operations & Supervision (OPS)

Field operations for supervised properties: readiness, inspections, tasks and incidents.

ID	Requirement	Priority	Status	Source
FR-OPS-01	The system shall allow assigning a Supervisor to supervised properties.	Must	Proposed	Client vision
FR-OPS-02	The system shall automatically create tasks: pre-arrival readiness before each check-in, and post-checkout inspection after each check-out.	Must	Proposed	Client vision
FR-OPS-03	The Supervisor app shall show today’s tasks with property, time, and navigation link.	Must	Proposed	Client vision
FR-OPS-04	The Supervisor shall complete a checklist per task, attach photos and notes, and mark the unit “Ready”.	Must	TBC	Client vision
FR-OPS-05	In supervised properties, a unit shall not become “Ready” for the next arrival until the readiness inspection is completed.	Must	TBC	Client vision
FR-OPS-06	The system shall allow reporting incidents and damage with severity, description and photos, linked to a booking and unit.	Must	TBC	Client vision
FR-OPS-07	The system shall provide a task board (To do / In progress / Done) for cleaning, readiness, inspection and maintenance tasks.	Should	Proposed	Analyst
FR-OPS-08	For self-service properties, the system shall support cleaning/maintenance tasks assigned to external service providers (mechanism TBC).	Should	TBC	Client vision

3.12 Content & Photography (MED)

Professional photography workflow and content quality review.

ID	Requirement	Priority	Status	Source
FR-MED-01	The system shall allow Admin/Operator to create a photography task for a property and assign a Photographer.	Should	Proposed	Client vision
FR-MED-02	The Photographer app shall show assigned shoots with address, time and shot list.	Should	Proposed	Client vision
FR-MED-03	The Photographer shall upload photos against the task and submit them for quality review.	Should	Proposed	Client vision
FR-MED-04	The Admin shall approve or reject each photo with a reason; rejected shoots return to the Photographer.	Should	Proposed	Client vision
FR-MED-05	Approved photos shall be added automatically to the property gallery.	Should	Proposed	Client vision

3.13 Smart Pricing (PRC)

AI pricing assistant that recommends nightly prices based on demand signals.

ID	Requirement	Priority	Status	Source
FR-PRC-01	The system shall generate a recommended nightly price per unit per date.	Must	Proposed	Client vision
FR-PRC-02	Recommendations shall consider public holidays, public events, special occasions, seasons, historical occupancy and lead time.	Must	Confirmed	Client vision
FR-PRC-03	The system shall show for each recommendation: current price, recommended price, direction (up/down), change % and the reason (signal).	Must	Proposed	Client vision
FR-PRC-04	The system shall allow the Operator to approve, reject or edit recommendations individually or in bulk.	Must	Proposed	Client vision
FR-PRC-05	The system shall support an optional auto-apply mode per property, constrained by Operator-defined minimum and maximum prices.	Should	TBC	Client vision
FR-PRC-06	The Admin shall maintain an events & holidays calendar used as pricing signals (e.g. Riyadh Season, Eid holidays, national day, school holidays).	Must	Proposed	Analyst
FR-PRC-07	The system shall support discounts (percentage off, with start/end date), displaying original and discounted price to guests.	Must	Confirmed	Client prototype

3.14 KPI Dashboards & Reporting (KPI)

Performance dashboards for Admin, Owners and Operators.

ID	Requirement	Priority	Status	Source
FR-KPI-01	The system shall provide KPI dashboards for Platform Admin (all properties), Owners (own properties) and Operators (assigned properties).	Must	Confirmed	Client vision
FR-KPI-02	The system shall compute: Bookings, Occupancy rate, Revenue, ADR, RevPAR, Cancellation rate, Average rating, Guest activity and Operational performance (see Section 9).	Must	Proposed	Client vision
FR-KPI-03	The system shall allow filtering KPIs by period, property, property type, city and operating model.	Must	Proposed	Client vision
FR-KPI-04	The system shall show trend charts and comparison to the previous period.	Should	Proposed	Analyst
FR-KPI-05	The system shall export reports to Excel (XLSX) and PDF.	Should	Recommended	Analyst
FR-KPI-06	KPIs shall be computed only from actual platform data using the approved definitions.	Must	Recommended	Client vision

3.15 Campaigns & Promotions (CMP)

Advertising campaigns and promotions with conversion tracking.

ID	Requirement	Priority	Status	Source
FR-CMP-01	The Admin shall be able to create and manage campaigns: name, goal, channel, budget, target properties, audience and dates.	Must	Confirmed	Client vision
FR-CMP-02	The system shall link each campaign to the properties it promotes (featured placement on home/search).	Should	Proposed	Client vision
FR-CMP-03	The system shall track campaign performance: impressions, visits, bookings, revenue and ROAS, using UTM parameters / promo codes.	Should	Proposed	Client vision

ID	Requirement	Priority	Status	Source
FR-CMP-04	The system shall support promo codes (percentage or fixed amount, usage limits, validity).	Should	Proposed	Analyst
FR-CMP-05	Owners shall be able to create unit-level promotions (discount %) subject to Admin rules (OQ-CMP-02).	Could	TBC	Client vision

3.16 Compliance & Government Integrations (INT)

Integrations with SHOMOOS, NAFATH and EWAA. Exact data contracts are TBC.

ID	Requirement	Priority	Status	Source
FR-INT-01	The system shall integrate with SHOMOOS to register guest (lodger) data with the competent security authority, triggered at confirmation or check-in (TBC).	Must	Confirmed	Client vision
FR-INT-02	The system shall integrate with NAFATH to verify guest/owner digital identity at registration, before booking or before key issuance (TBC).	Must	Confirmed	Client vision
FR-INT-03	The system shall integrate with EWAA; the target entity/system and scope (units, bookings or licences) must be confirmed before design.	Must	Confirmed	Client vision
FR-INT-04	The system shall log each integration call with status, response, timestamp and retry count, and retry failed calls automatically.	Must	Recommended	Client vision
FR-INT-05	The Admin shall see an integration health screen with last sync, error log and manual retry.	Should	Recommended	Analyst
FR-INT-06	The system shall block key issuance if a mandatory compliance step (e.g. SHOMOOS registration, if confirmed as mandatory) fails, and alert Operations.	Should	TBC	Analyst

3.17 Policies, Damage & Disputes (POL)

Damage & liability policy enforcement, incident review and fines.

ID	Requirement	Priority	Status	Source
FR-POL-01	The system shall display the damage & liability policy before booking confirmation.	Must	Proposed	Client vision
FR-POL-02	The system shall record the guest’s acknowledgement with date, time and policy version.	Must	Proposed	Client vision
FR-POL-03	The system shall allow opening a damage case from a post-stay inspection, linked to the booking, with photo evidence.	Must	Proposed	Client vision
FR-POL-04	The system shall route damage cases through a review & approval workflow (Opened → Under review → Approved / Rejected → Guest notified → Settled / Disputed).	Must	Proposed	Client vision
FR-POL-05	The system shall notify the guest of the case and its outcome and allow the guest to submit an objection.	Must	TBC	Client vision
FR-POL-06	The system shall apply the approved fine/claim by the approved mechanism (security deposit, card charge or wallet deduction — TBC).	Must	TBC	Client vision
FR-POL-07	No fine shall be applied without a documented review.	Must	TBC	Client vision

3.18 Notifications (NTF)

Transactional notifications across channels.

ID	Requirement	Priority	Status	Source
FR-NTF-01	The system shall send notifications via in-app, SMS, WhatsApp Business and e-mail, per user preferences.	Must	Proposed	Analyst
FR-NTF-02	The system shall notify guests of: booking confirmation, payment receipt, key issued, pre-arrival instructions, check-out reminder, review request, cancellation/refund, price alerts, damage case updates.	Must	Proposed	Client vision + Client prototype
FR-NTF-03	The system shall notify Owners/Operators of: new booking, cancellation, pricing recommendations, lock alerts, task updates, incidents, new reviews.	Must	Proposed	Analyst
FR-NTF-04	The system shall provide an in-app notification centre with read/unread state.	Should	Proposed	Analyst
FR-NTF-05	Notification templates shall be editable by Admin in Arabic and English.	Should	Recommended	Analyst

3.19 Financials & Payouts (FIN)

Revenue split between platform and owners, statements and payouts.

ID	Requirement	Priority	Status	Source
FR-FIN-01	The system shall calculate for each booking: gross amount, VAT, platform commission, payment fees, and owner net payout.	Must	TBC	Analyst
FR-FIN-02	The system shall generate periodic owner statements (monthly) with bookings, revenue, deductions and net payout.	Must	Proposed	Analyst
FR-FIN-03	The Admin shall be able to mark payouts as paid (bank transfer reference) and view outstanding balances.	Must	Proposed	Analyst
FR-FIN-04	The system shall record owner bank details (IBAN) verified before first payout.	Must	Recommended	Analyst

3.20 Platform Administration (ADM)

Back-office configuration and oversight.

ID	Requirement	Priority	Status	Source
FR-ADM-01	The Admin shall manage master data: cities, districts, property types, amenities.	Must	Proposed	Analyst
FR-ADM-02	The Admin shall manage commission %, VAT rate, cleaning fees, cancellation policy templates and loyalty settings.	Must	Proposed	Analyst
FR-ADM-03	The Admin shall review the property verification queue and approve/reject with reason.	Must	TBC	Client vision
FR-ADM-04	The Admin shall manage support tickets escalated by the assistant or raised by guests/owners.	Must	Proposed	Analyst
FR-ADM-05	The system shall keep an audit log of sensitive actions (role changes, price changes, refunds, fines, key issuance, data exports) with actor and timestamp.	Must	Recommended	Analyst
FR-ADM-06	The Admin shall manage CMS pages: Terms, Privacy Policy, FAQs, banners.	Should	Proposed	Client prototype

4. Key Use Cases

Main end-to-end scenarios. Detailed screen flows are demonstrated in the clickable prototype.

UC-01 — Register & sign in

Field	Description
Actors	Guest
Main flow	Guest enters mobile → receives OTP → enters OTP → (register) enters name, optional e-mail, password → account created, +100 points.
Alternative / exception	Invalid mobile format → inline error. Wrong OTP 3× → resend with cooldown.

UC-02 — Search & book a unit

Field	Description
Actors	Guest
Main flow	Search/filter → open unit → choose dates & guests → see price breakdown → verify identity (Nafath, if required) → acknowledge policy → pay (wallet / mada / card / Apple Pay) → confirmation with booking no. and digital key.
Alternative / exception	Dates unavailable → suggest alternatives. Insufficient wallet → top up or change method. Payment fails → dates released after 10 min.

UC-03 — Receive key & enter unit

Field	Description
Actors	Guest, Smart lock
Main flow	Booking confirmed → credential provisioned on lock → sent to guest → at check-in time guest enters PIN / taps “Open door” → lock opens → access logged.
Alternative / exception	Lock offline → fallback procedure; guest contacts assistant → ticket to Operator → emergency code.

UC-04 — Cancel booking

Field	Description
Actors	Guest
Main flow	Open booking → Cancel → system shows refund per policy → confirm → status Cancelled, key revoked, refund issued, cashback/points reversed, owner notified.
Alternative / exception	Outside free-cancellation window → partial/no refund per policy.

UC-05 — Onboard a property

Field	Description
Actors	Owner / Operator, Admin
Main flow	Add property wizard → upload documents → choose operating model → request photographer or upload photos → configure lock → submit → Admin verifies → published.
Alternative / exception	Verification rejected → owner notified with reason, can resubmit.

UC-06 — Review pricing recommendations

Field	Description
Actors	Operator

Field	Description
Main flow	Open Smart Pricing → calendar shows recommendations with reasons → approve / edit / reject → approved prices go live.
Alternative / exception	Auto-apply on → prices applied within min/max limits; operator notified.

UC-07 — Post-checkout inspection & damage

Field	Description
Actors	Supervisor, Admin, Guest
Main flow	Guest checks out → inspection task created → Supervisor completes checklist with photos → no damage: unit set to Preparing/Ready; damage: case opened → Admin reviews → guest notified → fine applied / case closed.
Alternative / exception	Guest objects → case moves to Disputed for Admin decision.

UC-08 — Photo shoot & QA

Field	Description
Actors	Photographer, Admin
Main flow	Shoot assigned → Photographer visits → uploads photos → Admin reviews each photo → approved photos published to gallery.
Alternative / exception	Rejected → retake requested with notes.

UC-09 — Compliance submission

Field	Description
Actors	Integration
Main flow	Booking confirmed / guest checked in → system sends guest & stay data to SHOMOOS → response stored → status visible in booking.
Alternative / exception	Failure → automatic retry with back-off → Admin alert after N failures.

UC-10 — Create & track campaign

Field	Description
Actors	Admin
Main flow	Create campaign → select properties, channel, budget, dates → generate UTM/promo code → campaign runs → dashboard shows visits, bookings, revenue, ROAS.
Alternative / exception	—

UC-11 — Wallet top-up & points redemption

Field	Description
Actors	Guest
Main flow	Wallet → choose amount → pay via gateway → balance updated. Points → Redeem → 100 points converted to 10 SAR in wallet.
Alternative / exception	Points < 100 → action disabled with message.

UC-12 — Request early check-in / late check-out

Field	Description
Actors	Guest, Operator
Main flow	Guest requests in active-stay screen → Operator approves/rejects → fee charged → key validity updated.

Field	Description
Alternative / exception	Conflicts with next booking → auto-reject with explanation.

5. Data Requirements

5.1 Core entities

Logical data model (indicative; the physical schema is defined in technical design).

Entity	Key attributes
User	id, role(s), name, mobile, email, password hash, nafath_verified, national ID / iqama (encrypted), language, status, created_at
Property	id, owner_id, type, name, city, district, geo location, operating_model (supervised/self-service), verification_status, listing_status, policies, created_at
Unit	id, property_id, name/code, capacity, bedrooms, amenities, base price, cleaning fee, operational_status, lock_id
Media	id, property_id/unit_id, url, order, is_cover, qa_status, photographer_id
Availability / Rate	unit_id, date, status (available/blocked/booked), price, min_stay, source (manual/recommended)
Booking	id, number, guest_id, unit_id, check_in, check_out, guests, nightly breakdown, fees, VAT, total, status, policy_ack_at, policy_version, created_at
Payment	id, booking_id, method, gateway_ref, amount, status, refunded_amount
Wallet / Transaction	wallet_id, user_id, balance; txn: id, type (top-up, payment, cashback, reward, refund, referral), amount, booking_id, description, created_at
Loyalty	user_id, points_balance, tier, points_ledger (earn/burn/expire), referral_code, referred_by
Access Credential	id, booking_id, unit_id, lock_id, code (encrypted), valid_from, valid_to, status (issued/active/expired/revoked)
Access Event	id, credential_id, lock_id, timestamp, result (success/fail), method
Lock	id, provider, external_id, unit_id, online, battery, last_seen
Task	id, type (readiness, inspection, cleaning, maintenance, photo shoot), unit_id, booking_id, assignee_id, due_at, status, checklist, photos, notes
Incident / Damage Case	id, booking_id, unit_id, reported_by, severity, description, photos, status, decision, amount, guest_objection
Review	id, booking_id, unit_id, guest_id, rating, comment, photos, owner_reply, visibility, created_at
Pricing Recommendation	id, unit_id, date, current_price, recommended_price, reason/signals, status (pending/approved/rejected/auto)
Event / Holiday	id, name, city, start, end, impact level
Campaign / Promo code	id, name, channel, budget, dates, properties, UTM, code, discount, usage limit, metrics
Integration Log	id, system (SHOMOOS/NAFATH/EWAA/payment/lock/SMS), entity ref, request, response, status, retries, timestamp
Support Ticket / Conversation	id, user_id, booking_id, channel (assistant/app/WhatsApp), category, status, messages
Payout / Statement	id, owner_id, period, gross, commission, fees, net, status, bank_ref
Audit Log	id, actor_id, action, entity, before/after, ip, timestamp

5.2 State models

Object	States & transitions
Booking	Pending Payment → Confirmed → Checked-in → Completed; Confirmed → Cancelled; Pending Payment → Expired (hold timeout); Confirmed → No-show

Object	States & transitions
Property listing	Draft → Submitted → Under Verification → Approved/Published → Suspended; Under Verification → Rejected → Draft
Unit operational status	Available → Booked → Occupied → Post-checkout Inspection → Preparing → Ready/Available; any → Blocked/Maintenance
Access credential	Issued → Active (at valid_from) → Expired (at valid_to); Issued/Active → Revoked (cancellation/security)
Damage case	Opened → Under Review → Approved / Rejected → Guest Notified → Settled | Disputed → Final Decision → Closed
Pricing recommendation	Pending → Approved | Edited & Approved | Rejected | Auto-applied | Expired (date passed)
Task	To Do → In Progress → Done | Blocked; photo shoot: Submitted → QA Approved | Retake Requested

5.3 Property profile — five data layers

Layer	Fields
Identity	Property information · property type · location
Content	Photos · amenities · description
Commercial	Availability · pricing · booking status
Operations	Operational status · supervision type · access settings
Governance	Policies · guest data · verification documents

6. External Interface Requirements

6.1 User interface — Rawaaf design system

The client supplied a working Rawaaf guest prototype. All portals shall reuse its visual language so the product feels like one brand:

Token	Value	Usage
Primary / Navy	#0A1931	Text, primary buttons, active nav, dark hero cards
Accent / Gold	#C9A86A (text #8B6F47)	Highlights, CTAs on dark, progress, tier badges
Background / Cream	#F7F5F0	Page background, input fills, secondary chips
Surface	#FFFFFF, radius 20–24 px, soft shadow	Cards, panels, modals
Status colours	Emerald = confirmed · Navy/Blue = completed · Red = cancelled · Amber = attention	Status pills
Typography	Tajawal 300–800 (Arabic & Latin); heavy 800 headings	All text
Shape	Pill buttons & chips (fully rounded); 64 px sticky blurred header	Controls, navigation
Direction	RTL default, LTR for English	Layout

UX principles: mobile-first; booking in ≤ 5 steps; price transparency before payment (all fees + VAT); status always visible (pills); every destructive action confirmed; empty states with a next action; toasts instead of browser alerts; skeleton loading.

6.2 Screen inventory

Each screen below is represented in the clickable prototype (prototype/index.html), which is the UI reference for this SRS.

Portal	Screens	Prototype routes
Guest website / app	Home (hero search, city & type filters, unit grid, compare tray) · Search results · Unit detail (gallery, details, amenities, policies, reviews) · Compare modal · Booking wizard (5 steps incl. Nafath & policy acknowledgement) · Confirmation & digital key · Login / Register with OTP · Account: overview, bookings, favourites, wallet, points & rewards, reviews, settings · Active stay (open door, Wi-Fi, guide, requests, report issue, check-out) · Review · Invoice · Assistant chat · Notifications	#/ #/search #/property/:id #/book/:id #/account/:tab #/stay/:id #/review/:id #/invoice/:id
Owner / Operator dashboard	KPI dashboard · Properties list · Property profile (5 layers) · Add-property wizard (9 steps) · Availability calendar · Smart pricing · Bookings list & detail · Smart locks & access events · Operations: task board, incidents, damage cases · Reviews & replies · Payouts · Promotions	#/operator …/properties …/property/:id/:tab …/add …/calendar …/pricing …/bookings …/booking/:id …/locks …/ops/:tab …/reviews …/payouts …/campaigns
Supervisor app	Today’s tasks · Task detail with checklist & photos · Mark unit ready · Report incident / damage · Map · Reports · Profile	#/supervisor …/task/:id …/map …/reports …/me
Photographer app	Assigned shoots · Shoot detail & shot list · Upload photos · QA feedback · Earnings · Profile	#/photographer …/shoot/:id …/earnings …/me

Portal	Screens	Prototype routes
Platform Admin console	Platform KPIs · Property verification queue · Photo QA · Damage & disputes · Users & roles (permission matrix) · Campaigns · Integrations health · Pricing signals & events · Settings (master data, fees, VAT, policies, loyalty) · Audit log	#/admin …/verification …/photos …/disputes …/users …/campaigns …/integrations …/pricing …/settings …/audit

Open prototype/index.html in a browser; use the floating "Demo" button to switch portals. Guest OTP for the demo is 1234.

6.3 Hardware interfaces

- Smart locks (keypad PIN, Bluetooth e-key and/or remote unlock via gateway), with battery and connectivity telemetry.
- Mobile device camera for inspection and photography uploads.

6.4 Software interfaces

System	Purpose	Notes / constraints
Payment gateway	Card/mada/Apple Pay payments, tokenisation, refunds, webhooks.	Provider TBC (e.g. Moyasar, HyperPay, Tap, PayTabs). Must be SAMA-licensed and PCI-DSS compliant.
NAFATH	Identity verification (push request + number matching) returning verified identity attributes.	Via the approved national integration channel; scope TBC.
SHOMOOS	Registration of lodger data for accommodation facilities.	Data fields, trigger and response TBC.
EWAA	Accommodation-sector integration identified by client.	Target entity/system and scope TBC.
Smart-lock provider	Provision/revoke PINs or e-keys, remote unlock, lock status & battery, access events (webhooks).	Provider TBC (e.g. TTLock, Nuki, igloohome). Offline-code support recommended.
SMS / OTP	OTP and transactional SMS.	Saudi-registered sender ID required (CST regulations).
WhatsApp Business API	Transactional messages and support.	Pre-approved templates required.
E-mail	Transactional e-mail.	SPF/DKIM configured domain.
Maps	Geocoding, map display, directions.	Google Maps or equivalent.
AI / LLM provider	Personal assistant and pricing explanations.	Model/provider TBC; data-residency and PDPL review required.
ZATCA (Fatoora)	E-invoice generation/reporting.	Phase 2 integration requirements apply to the invoicing entity.
Analytics / Ads	Campaign tracking (UTM, pixels).	Channels TBC.

6.5 Communication interfaces

- HTTPS/TLS 1.2+ REST/JSON APIs; webhooks for payment, lock and integration callbacks (signed and idempotent).
- Push notifications (web push / mobile), SMS, WhatsApp Business, e-mail.

7. Non-Functional Requirements

Category	ID	Requirement
Performance	NFR-PER-01	Guest pages shall load (Largest Contentful Paint) in ≤ 2.5 s on a 4G connection for the 75th percentile.
Performance	NFR-PER-02	Search and availability API responses shall be ≤ 800 ms at p95 under expected load.
Performance	NFR-PER-03	A digital key shall be provisioned to the lock within 60 s of booking confirmation (when the lock is online).
Scalability	NFR-SCA-01	The system shall support at least 5,000 units, 1,000 concurrent users and 10× peak during events/holidays without redesign (horizontal scaling).
Availability	NFR-AVA-01	Guest booking and smart-access services shall have 99.9% monthly availability; planned maintenance outside 00:00–06:00 KSA is excluded.
Reliability	NFR-REL-01	Backups: daily full + point-in-time recovery; RPO ≤ 15 min, RTO ≤ 4 h.
Reliability	NFR-REL-02	Smart access shall remain usable when the platform is down (codes are pre-provisioned on the lock / offline codes).
Security	NFR-SEC-01	All traffic over TLS 1.2+; data at rest encrypted (AES-256); access codes and ID numbers encrypted at field level.
Security	NFR-SEC-02	Card data shall never be stored or transit platform servers (gateway tokenisation / hosted fields) — PCI-DSS SAQ-A scope.
Security	NFR-SEC-03	OWASP Top-10 and OWASP ASVS L2 controls; annual penetration test and before go-live.
Security	NFR-SEC-04	Rate limiting on authentication, OTP and key-reveal endpoints; all admin actions audited.
Privacy	NFR-PRV-01	Comply with the Saudi Personal Data Protection Law (PDPL) and SDAIA regulations: consent, purpose limitation, data-subject rights, breach notification.
Privacy	NFR-PRV-02	Personal and compliance data shall be hosted in Saudi Arabia (data residency), unless an approved exception applies.
Privacy	NFR-PRV-03	Defined retention per data type (IDs, access logs, inspection photos, conversations) — values TBC with legal counsel.
Localisation	NFR-LOC-01	Arabic (RTL, default) and English (LTR); currency SAR; Gregorian dates with optional Hijri display; KSA time zone (UTC+3).
Usability	NFR-USA-01	UI shall follow the Rawaaf design system (Section 6.1) consistently across portals; responsive from 360 px to 1440 px+.
Accessibility	NFR-ACC-01	WCAG 2.1 AA for guest-facing screens (contrast, keyboard navigation, labels, focus states).
Compatibility	NFR-CMP-01	Latest two versions of Chrome, Safari, Edge, Firefox; iOS 15+ and Android 9+ for mobile web/app.
Maintainability	NFR-MNT-01	Modular architecture with documented REST APIs (OpenAPI); automated tests ≥ 70% coverage on core business logic; CI/CD with staging environment.
Observability	NFR-OBS-01	Centralised logging, metrics and alerting for payments, integrations and lock events.
Auditability	NFR-AUD-01	Immutable audit trail for financial, access and compliance events, retained ≥ 5 years (TBC).

8. Business Rules

ID	Rule	Status	Source
BR-001	An access credential is issued only for a valid, confirmed (paid) booking.	Confirmed	Client vision
BR-002	A credential is bound to one booking and one unit only.	Confirmed	Client vision
BR-003	Unit price may vary according to defined demand factors (holidays, events, occasions, seasons).	Confirmed	Client vision
BR-004	Each property has one operating model: Supervised or Self-service.	Confirmed	Client vision
BR-005	A unit cannot be booked for dates where it is unavailable (no double booking).	Proposed	Client vision
BR-006	A credential is valid only during the stay window (default 15:00 arrival – 12:00 departure).	TBC	Client vision + Client prototype
BR-007	A credential is revoked when its booking is cancelled.	Proposed	Client vision
BR-008	The guest must acknowledge the property policy before booking confirmation.	TBC	Client vision
BR-009	No fine is applied without a documented incident review.	TBC	Client vision
BR-010	Applying a recommended price requires Operator approval unless auto-apply is enabled within min/max limits.	TBC	Client vision
BR-011	A property is not published before mandatory data, photos and verification are complete.	Proposed	Client vision
BR-012	In supervised properties a unit is not “Ready” before the readiness inspection is completed.	TBC	Client vision
BR-013	Post-stay inspection is completed before the unit is released for the next booking.	TBC	Client vision
BR-014	Every access attempt, successful or failed, is logged with time and unit.	Recommended	Client vision
BR-015	Identity verification may be required before booking or before key issuance.	TBC	Client vision
BR-016	Guest data is sent to competent authorities according to approved regulatory requirements.	TBC	Client vision
BR-017	The assistant never performs financial actions or booking changes without approved permission and guest confirmation.	Recommended	Client vision
BR-018	KPIs are computed only from actual platform data with approved definitions.	Recommended	Client vision
BR-019	Wallet payment is allowed only if balance ≥ booking total (no partial wallet payment in MVP).	Confirmed	Client prototype
BR-020	Cashback is credited to the wallet at the guest’s tier rate after payment and reversed if the booking is cancelled.	Confirmed	Client prototype
BR-021	Points: earn 10 per 100 SAR paid; redeem 100 points = 10 SAR; minimum redemption 100 points.	Confirmed	Client prototype
BR-022	Only guests with a Completed booking for the unit may review it; one review per booking.	Confirmed	Client prototype
BR-023	Referral reward (50 SAR each) is granted only after the referred user’s first completed/paid booking.	Confirmed	Client prototype
BR-024	A guest may compare up to 3 units at a time.	Confirmed	Client prototype
BR-025	Mobile numbers must start with 05 and be 10 digits.	Confirmed	Client prototype

9. KPI Definitions

Proposed standard definitions, to be approved with the client. Values are computed only from actual platform data.

KPI	Meaning	Formula
Bookings	Count of confirmed bookings in the period.	Count(status ∈ {Confirmed, Checked-in, Completed})
Occupancy rate	Share of available nights that were sold.	Nights sold ÷ Nights available × 100
Revenue	Accommodation revenue in the period (excl. VAT).	Σ booking net accommodation amount
ADR	Average Daily Rate.	Accommodation revenue ÷ Nights sold
RevPAR	Revenue per available unit-night.	Accommodation revenue ÷ Nights available (= ADR × Occupancy)
Cancellation rate	Share of bookings cancelled.	Cancelled bookings ÷ Total bookings × 100
Average rating	Guest satisfaction.	Mean of review ratings in the period
Guest activity	Engagement & loyalty.	Active guests, repeat-booking rate, assistant conversations
Operational performance	Field operations quality.	Turnover time between stays, inspection pass rate, incidents per 100 stays, lock uptime
Campaign ROAS	Return on ad spend.	Attributed revenue ÷ Campaign spend

Analysis dimensions: period · property · property type · city · operating model.

10. Open Questions & Decisions Required

These items block detailed design of the related requirements (Status = TBC). Please answer or confirm each one.

ID	Area	Question	Client answer
OQ-PRP-01	PRP	Who may list a property on the platform (individuals, companies, licensed operators only)?	
OQ-PRP-02	PRP	Who verifies ownership/operator authorisation, and which documents are required (title deed, lease, tourism licence)?	
OQ-PRP-03	PRP	Which property types are in scope for launch?	
OQ-PRP-04	PRP	Are hotels managed as independent units or as one facility with multiple rooms?	
OQ-BKG-01	BKG	Is online payment mandatory, or is pay-at-property allowed?	
OQ-BKG-02	BKG	What are the cancellation rules (flexible / moderate / strict templates, fees)?	
OQ-BKG-03	BKG	What happens on booking modification (price difference, fees)?	
OQ-PAY-01	PAY	Which payment gateway and payment methods (mada, Apple Pay, STC Pay, Tabby/Tamara BNPL)?	
OQ-PAY-02	PAY	Is a security deposit required? How is it held and released?	
OQ-LOY-01	LOY	Client site shows flat 5% cashback but tier table shows 3% / 5% / 7%. Which is correct?	
OQ-LOY-02	LOY	Do points or wallet balance expire? Can wallet balance be withdrawn to a bank account?	
OQ-ACC-01	ACC	Which smart-lock providers will be supported?	
OQ-ACC-02	ACC	Are codes generated by the platform or by the lock provider system?	
OQ-ACC-03	ACC	When exactly does key validity start and end? Are check-in/out times per property?	
OQ-ACC-04	ACC	What is the fallback when a lock is offline or fails?	
OQ-PRC-01	PRC	Is smart pricing advisory or automatic?	
OQ-PRC-02	PRC	Which demand signals must affect price (and data sources for events)?	
OQ-PRC-03	PRC	Who approves price changes (owner vs operator)?	
OQ-OPS-01	OPS	What exactly does the Supervisor manage? Final checklist content?	
OQ-OPS-02	OPS	What is the photographer workflow? Is photography paid by the owner?	
OQ-OPS-03	OPS	Are supervisors and photographers employees or external partners?	
OQ-OPS-04	OPS	Who handles cleaning in self-service properties?	
OQ-INT-01	INT	Exact data to exchange with SHOMOOS and trigger point?	

ID	Area	Question	Client answer
OQ-INT-02	INT	Role of NAFATH in the journey: registration, booking, or key issuance? For Saudis only or also residents/GCC/visitors?	
OQ-INT-03	INT	Which entity/system is meant by EWAA, and what data is exchanged?	
OQ-POL-01	POL	How is damage verified? Who approves fines? How is the guest notified and can they object?	
OQ-POL-02	POL	Fine amounts, legal wording and collection mechanism?	
OQ-AST-01	AST	Which data can the assistant access?	
OQ-AST-02	AST	Can the assistant modify or cancel bookings?	
OQ-AST-03	AST	Can it contact external systems? Which languages must it support?	
OQ-CMP-01	CMP	Which advertising channels are targeted?	
OQ-CMP-02	CMP	Who manages campaigns: the platform or the owner?	
OQ-BUS-01	BUS	Is the first launch web, mobile app (iOS/Android), or both?	
OQ-BUS-02	BUS	What is the platform revenue model: commission %, subscription, or other? Who is the merchant of record for VAT invoices?	
OQ-BUS-03	BUS	The client prototype mentions “units from the other competitor are not available here” — is there a multi-brand / white-label requirement?	
OQ-BUS-04	BUS	Is English required at launch, or Arabic only?	

11. Proposed Release Plan

A logical build sequence for discussion — not an agreed timeline.

Phase	Content
Phase 1 — Core booking	Accounts & OTP, discovery & search, property/unit management, availability, booking wizard, payment gateway, wallet & invoices, reviews, guest account, basic admin.
Phase 2 — Smart access	Smart-lock integration, key issuance/revocation, access log, active-stay screen, lock monitoring, fallback procedure.
Phase 3 — Operations & KPIs	Supervisor & photographer apps, tasks, inspections, incidents & damage cases, KPI dashboards, campaigns, owner statements & payouts.
Phase 4 — Compliance	SHOMOOS, NAFATH, EWAA integrations and integration monitoring. May move to Phase 1 if legally mandatory before operating.
Phase 5 — Intelligence	Smart pricing recommendations, personal AI assistant, loyalty tiers & referrals (if not in Phase 1).
Phase 6 — Advanced (future)	Demand forecasting, personalised guest experience, proactive operational insights.

12. Requirements Summary & Sign-off

Measure	Count
Functional requirements	160
Must / Should / Could	111 / 43 / 6
Confirmed / Proposed / Recommended / TBC	58 / 55 / 24 / 23
Non-functional requirements	21
Business rules	25
Use cases	12
Open questions	36

By signing below, the parties confirm that this SRS (with answers to Section 10) is the baseline for design and development. Changes after sign-off follow the change-request process.

Role	Name	Signature	Date
Client — Product owner			
Client — Business representative			
Delivery — Business analyst			
Delivery — Technical lead			